🏆 Infosys SE Elite Crack Vault — 02 Multiplication Division Factor and Exponential Puzzles Elite

🎯 Infosys Systems Engineer (SE) | 📊 10 Questions | 🗓️ 5-6 Nov 2026 Drive | ⚡ 100% Pure SE Verified

Problem 1 • Matrix Number Puzzle 16

🔥 MUST-SOLVE

ID: Q1 | Source: Infosys PYQ | Level: Infosys SE Standard

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
6	3	2
64	8	8
63	9	?

- [ ] **A)** 17
- [ ] **B)** 10
- [x] **C)** 7
- [ ] **D)** 25

✓ **Verified Answer:** ### Correct Answer Option C) 7** #### Detailed Mathematical Derivation 1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3. 2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 63 ÷ 9 = 7. 3. **Step-by-Step Algebraic Verification:** - Evaluating the operations across corresponding rows and columns confirms this uniform pattern. - Applying the identical transformation to the row/column containing the target cell produces True = 7$. 4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 7. #### Distractor & Elimination Analysis - **Option A (17):** Incorrect. Violates the established matrix row/column transformation rule. - **Option B (10):** Incorrect. Violates the established matrix row/column transformation rule. - **Option C (7):** Correct. Exactly matches the derived mathematical rule with zero contradiction. - **Option D (25):** Incorrect. Violates the established matrix row/column transformation rule.

Problem 2 • Matrix Number Puzzle 17

MUST-SOLVE

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
15	3	5
36	6	6
8	2	?

- [ ] **A)** 12
- [ ] **B)** 8
- [ ] **C)** 19
- [x] **D)** 4

✓ **Verified Answer:** ### Correct Answer Option D) 4** #### Detailed Mathematical Derivation 1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3. 2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 8 ÷ 2 = 4. 3. **Step-by-Step Algebraic Verification:** - Evaluating the operations across corresponding rows and columns confirms this uniform pattern. - Applying the identical transformation to the row/column containing the target cell produces True = 4$. 4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 4. #### Distractor & Elimination Analysis - **Option A (12):** Incorrect. Violates the established matrix row/column transformation rule. - **Option B (8):** Incorrect. Violates the established matrix row/column transformation rule. - **Option C (19):** Incorrect. Violates the established matrix row/column transformation rule. - **Option D (4):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Problem 3 • Matrix Number Puzzle 18

MUST-SOLVE

ID: Q3 | Source: Infosys PYQ | Level: Infosys SE Standard

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
6	2	3
8	4	2
72	8	?

- [ ] **A)** 6
- [ ] **B)** 13
- [x] **C)** 9
- [ ] **D)** 3

✓ **Verified Answer:** ### Correct Answer Option C) 9** #### Detailed Mathematical Derivation 1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3. 2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 72 ÷ 8 = 9. 3. **Step-by-Step Algebraic Verification:** - Evaluating the operations across corresponding rows and columns confirms this uniform pattern. - Applying the identical transformation to the row/column containing the target cell produces True = 9$. 4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 9. #### Distractor & Elimination Analysis - **Option A (6):** Incorrect. Violates the established matrix row/column transformation rule. - **Option B (13):** Incorrect. Violates the established matrix row/column transformation rule. - **Option C (9):** Correct. Exactly matches the derived mathematical rule with zero contradiction. - **Option D (3):** Incorrect. Violates the established matrix row/column transformation rule.

Problem 4 • Matrix Number Puzzle 19

CORE SE

ID: Q4 | Source: Infosys PYQ | Level: Infosys SE Standard

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
12	6	2
36	6	6
8	4	?

- [x] **A)** 2
- [ ] **B)** 6
- [ ] **C)** 5
- [ ] **D)** 7

✓ **Verified Answer:** ### Correct Answer Option A) 2** #### Detailed Mathematical Derivation 1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3. 2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 8 ÷ 4 = 2. 3. **Step-by-Step Algebraic Verification:** - Evaluating the operations across corresponding rows and columns confirms this uniform pattern. - Applying the identical transformation to the row/column containing the target cell produces True = 2$. 4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 2. #### Distractor & Elimination Analysis - **Option A (2):** Correct. Exactly matches the derived mathematical rule with zero contradiction. - **Option B (6):** Incorrect. Violates the established matrix row/column transformation rule. - **Option C (5):** Incorrect. Violates the established matrix row/column transformation rule. - **Option D (7):** Incorrect. Violates the established matrix row/column transformation rule.

Problem 5 • Matrix Number Puzzle 20

CORE SE

ID: Q5 | Source: Infosys PYQ | Level: Infosys SE Standard

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
15	5	3
49	7	7
42	6	?

- [ ] **A)** 22
- [x] **B)** 7
- [ ] **C)** 12
- [ ] **D)** 4

✓ **Verified Answer:** ### Correct Answer Option B) 7** #### Detailed Mathematical Derivation 1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3. 2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 42 ÷ 6 = 7. 3. **Step-by-Step Algebraic Verification:** - Evaluating the operations across corresponding rows and columns confirms this uniform pattern. - Applying the identical transformation to the row/column containing the target cell produces True = 7$. 4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 7. #### Distractor & Elimination Analysis - **Option A (22):** Incorrect. Violates the established matrix row/column transformation rule. - **Option B (7):** Correct. Exactly matches the derived mathematical rule with zero contradiction. - **Option C (12):** Incorrect. Violates the established matrix row/column transformation rule. - **Option D (4):** Incorrect. Violates the established matrix row/column transformation rule.

Problem 6 • Matrix Number Puzzle 21

CORE SE

ID: Q6 | Source: Infosys PYQ | Level: Infosys SE Standard

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
5	3	28
5	1	26
1	7	?

- [x] **A)** 8
- [ ] **B)** 14
- [ ] **C)** 18
- [ ] **D)** 12

✓ **Verified Answer:** ### Correct Answer Option A) 8** #### Detailed Mathematical Derivation 1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3. 2. **Governing Rule & Logic:** Each row follows: third = first^2 + second. So ? = 1^2 + 7 = 8. 3. **Step-by-Step Algebraic Verification:** - Evaluating the operations across corresponding rows and columns confirms this uniform pattern. - Applying the identical transformation to the row/column containing the target cell produces True = 8$. 4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 8. #### Distractor & Elimination Analysis - **Option A (8):** Correct. Exactly matches the derived mathematical rule with zero contradiction. - **Option B (14):** Incorrect. Violates the established matrix row/column transformation rule. - **Option C (18):** Incorrect. Violates the established matrix row/column transformation rule. - **Option D (12):** Incorrect. Violates the established matrix row/column transformation rule.

Problem 7 • Matrix Number Puzzle 22

CORE SE

ID: Q7 | Source: Infosys PYQ | Level: Infosys SE Standard

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
7	5	54
6	10	46
7	8	?

- [ ] **A)** 49
- [ ] **B)** 54
- [x] **C)** 57
- [ ] **D)** 45

✓ **Verified Answer:** ### Correct Answer Option C) 57** #### Detailed Mathematical Derivation 1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3. 2. **Governing Rule & Logic:** Each row follows: third = first^2 + second. So ? = 7^2 + 8 = 57. 3. **Step-by-Step Algebraic Verification:** - Evaluating the operations across corresponding rows and columns confirms this uniform pattern. - Applying the identical transformation to the row/column containing the target cell produces True = 57$. 4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 57. #### Distractor & Elimination Analysis - **Option A (49):** Incorrect. Violates the established matrix row/column transformation rule. - **Option B (54):** Incorrect. Violates the established matrix row/column transformation rule. - **Option C (57):** Correct. Exactly matches the derived mathematical rule with zero contradiction. - **Option D (45):** Incorrect. Violates the established matrix row/column transformation rule.

Problem 8 • Matrix Number Puzzle 23

CORE SE

ID: Q8 Source: Infosys PYQ Level: Infosys SE Standard

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
6	5	41
6	1	37
5	2	?

- [ ] **A)** 37
- [ ] **B)** 12
- [x] **C)** 27
- [ ] **D)** 42

✓ **Verified Answer:** ### Correct Answer Option C) 27** #### Detailed Mathematical Derivation 1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3. 2. **Governing Rule & Logic:** Each row follows: third = first^2 + second. So ? = 5^2 + 2 = 27. 3. **Step-by-Step Algebraic Verification:** - Evaluating the operations across corresponding rows and columns confirms this uniform pattern. - Applying the identical transformation to the row/column containing the target cell produces True = 27$. 4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 27. #### Distractor & Elimination Analysis - **Option A (37):** Incorrect. Violates the established matrix row/column transformation rule. - **Option B (12):** Incorrect. Violates the established matrix row/column transformation rule. - **Option C (27):** Correct. Exactly matches the derived mathematical rule with zero contradiction. - **Option D (42):** Incorrect. Violates the established matrix row/column transformation rule.

Problem 9 • Matrix Number Puzzle 24

CORE SE

ID: Q9 Source: Infosys PYQ Level: Infosys SE Standard

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
3	5	14
5	4	29
5	2	?

- [ ] **A)** 15
- [x] **B)** 27
- [ ] **C)** 21
- [ ] **D)** 23

✓ **Verified Answer:** ### Correct Answer Option B) 27** #### Detailed Mathematical Derivation 1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3. 2. **Governing Rule & Logic:** Each row follows: third = first^2 + second. So ? = 5^2 + 2 = 27. 3. **Step-by-Step Algebraic Verification:** - Evaluating the operations across corresponding rows and columns confirms this uniform pattern. - Applying the identical transformation to the row/column containing the target cell produces True = 27$. 4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 27. #### Distractor & Elimination Analysis - **Option A (15):** Incorrect. Violates the established matrix row/column transformation rule. - **Option B (27):** Correct. Exactly matches the derived mathematical rule with zero contradiction. - **Option C (21):** Incorrect. Violates the established matrix row/column transformation rule. - **Option D (23):** Incorrect. Violates the established matrix row/column transformation rule.

Problem 10 • Matrix Number Puzzle 25

CORE SE

ID: Q10 | Source: Infosys PYQ | Level: Infosys SE Standard

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
1	2	3
6	5	41
7	5	?

- [ ] **A)** 66
- [x] **B)** 54
- [ ] **C)** 49
- [ ] **D)** 39

✓ **Verified Answer:** ### Correct Answer Option B) 54** #### Detailed Mathematical Derivation 1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3. 2. **Governing Rule & Logic:** Each row follows: third = first^2 + second. So ? = 7^2 + 5 = 54. 3. **Step-by-Step Algebraic Verification:** - Evaluating the operations across corresponding rows and columns confirms this uniform pattern. - Applying the identical transformation to the row/column containing the target cell produces True = 54$. 4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 54. #### Distractor & Elimination Analysis - **Option A (66):** Incorrect. Violates the established matrix row/column transformation rule. - **Option B (54):** Correct. Exactly matches the derived mathematical rule with zero contradiction. - **Option C (49):** Incorrect. Violates the established matrix row/column transformation rule. - **Option D (39):** Incorrect. Violates the established matrix row/column transformation rule.